

Chemical All Worksheets Formulas

Chapter 7 Part I

Types of Chemical Compounds

Classify the following compounds as ionic or molecular (covalent).

- | | |
|--------------------------------------|--|
| 1. CaCl_2 _____ ionic _____ | 11. MgO _____ |
| 2. CO_2 _____ | 12. NH_4Cl _____ |
| 3. H_2O _____ | 13. $\text{Sr}(\text{NO}_3)_2$ _____ |
| 4. Na_2SO_4 _____ | 14. KI _____ |
| 5. K_2O _____ | 15. $\text{Ba}(\text{OH})_2$ _____ |
| 6. NaF _____ | 16. NO_2 _____ |
| 7. Na_2CO_3 _____ | 17. $\text{Ca}_3(\text{PO}_4)_2$ _____ |
| 8. CH_4 _____ | 18. FeCl_3 _____ |
| 9. $\text{Mg}(\text{NO}_3)_2$ _____ | 19. P_2O_5 _____ |
| 10. LiBr _____ | 20. N_2O_3 _____ |

Binary Ionic Compounds – Compounds with monatomic ions in it, a metallic ion and a nonmetallic ion. This allows only two types of atoms in the formula. Ex: Rb_2O

Ternary Ionic Compounds – Compounds with at least one polyatomic ion in it. This allows three or more types of atoms in the formula. Ex: RbNO_3

Classify the following compounds as binary ionic or ternary ionic.

- | | |
|--|--|
| 21. KOH _____ ternary ionic _____ | 26. $\text{Na}_2\text{Cr}_2\text{O}_7$ _____ |
| 22. CoO _____ | 27. MgSO_4 _____ |
| 23. $\text{Fe}(\text{NO}_3)_2$ _____ | 28. Cu_2S _____ |
| 24. MgH_2 _____ | 29. SnO_2 _____ |
| 25. Cs_2S _____ | 30. NH_4NO_3 _____ |

Naming Ions

Polyatomic Ions

Rule: Look up the name or symbol from your polyatomic ion sheet.

Example 1: ammonium ion _____

Example 2: NO_3^- _____

Example 3: NO_2^- _____

Monatomic Ions

• CATIONS

Rule A: If the atom always forms the same charge when forming an ion, (all group 1, group 2, and Zn^{2+} , Ag^+ , Cd^{2+} , & Al^{3+}), take the name of the atom that the ion is formed from, and add “ion”.

Example 4: Na^+ _____

Example 5: Mg^{2+} _____

Example 6: aluminum ion _____

Rule B: If the atom can form more than one charge when forming an ion, (any of the transition metals and any metals underneath the staircase), take the name of the atom that the ion is formed from, place the charge as a Roman numeral in parentheses, and then add “ion”.

Example 7: Pb^{2+} _____

Example 9: copper(I) ion _____

Example 8: Fe^{2+} _____

Example 10: copper(II) ion _____

Try These:

11. Rb^+ _____

15. Fe^{3+} _____

12. Ca^{2+} _____

16. cobalt(II) ion _____

13. Ni^+ _____

17. lithium ion _____

14. Ag^+ _____

18. zinc ion _____

- ANIONS

Rule: Take the nonmetal atom name, remove the ending and add “-ide ion” to it.

Example 21: S^{2-} _____

Example 22: N^{3-} _____

Example 23: bromide ion _____

Example 24: telluride ion _____

YOU TRY IT!

25. iodide ion _____

26. selenide ion _____

27. F^- _____

28. O^{2-} _____

Let's Compare some ions:

N^{3-} _____

NO_2^- _____

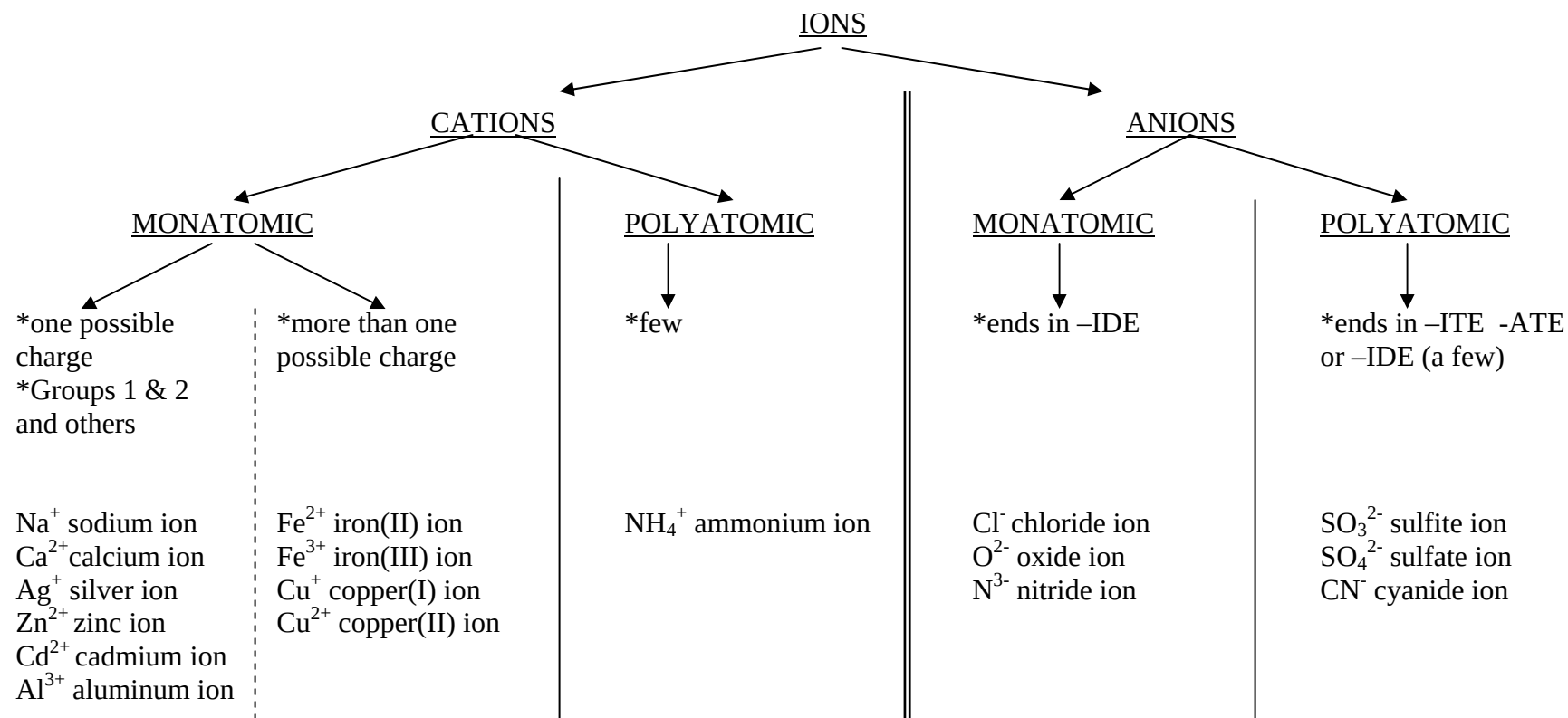
NO_3^- _____

sulfide ion _____

sulfite ion _____

sulfate ion _____

Understanding Ion Nomenclature



Monatomic cation with one possible charge – Name of atom and add “ion”

Monatomic cation with more than one possible charge – Name of atom, with charge as Roman numeral in parentheses, and add “ion”

Monatomic anion – Name of atom, remove ending, and add “-ide ion”

Polyatomic ions – no naming rules...just know them.

Ions - Worksheet

Name the following ions.

1. Ca^{2+} _____

2. O^{2-} _____

3. H^+ _____

4. H^- _____

5. Cu^+ _____

6. Fe^{3+} _____

7. CO_3^{2-} _____

8. NH_4^+ _____

9. Zn^{2+} _____

10. N^{3-} _____

Write the formulas for the following ions.

11. acetate ion _____

12. phosphide ion _____

13. phosphate ion _____

14. iron(II) ion _____

15. strontium ion _____

16. nickel(II) ion _____

17. tin(II) ion _____

18. sulfate ion _____

19. sulfite ion _____

20. sulfide ion _____

Ionic Compound Nomenclature

ALL COMPOUNDS ARE NEUTRAL!

When ions combine, they will form neutral compounds. These formulas are written in the lowest, whole-number ratio. These lowest, whole-number ratios are called “formula units”.

An ionic compound is often referred to as a “salt”. One of the most common ionic compounds is NaCl. This is why we usually refer to it as salt or table salt.

When determining the formula for an ionic compound, determine the charges of each ion and be sure to combine them so that the number of positive charges is equal in magnitude to the number of negative charges.

Example: Al_2O_3 - aluminum oxide

It is made of two Al^{3+} ions (aluminum ions) and three O^{2-} ions (oxide ions).

2 Al^{3+} ions would make a total positive charge of +6.

3 O^{2-} ions would make a total negative charge of -6.

This makes a neutral compound.

Ionic Compounds are named after the ions they contain.

Example: **CaI_2**

This contains 1 Ca^{2+} ion and 2 I^- ions. They are written in a 1:2 ratio in the compound so it is neutral. However, when naming the compound, just write the names of the two types of ions. So the name is **calcium iodide**.

Example: **MnO_2**

This contains 1 Mn^{+4} ion and 2 O^{2-} ions. I had to determine the charge on the Mn. It is a transition metal and needs a roman numeral in its name. I first looked at the O and knew it was a 2-. I then thought that if there were two 2- ions, then that would be a total negative charge of 4-. In order to make a neutral compound, the positives would have to add up to 4+. If there is only one Mn in the compound, then its entire charge would be 4+. So, the compound is made of manganese(IV) ions and oxide ions. The name is **manganese (IV) oxide**.

Example: copper (I) sulfide

This contains Cu^+ ions and S^{2-} ions. Two Cu^+ ions are needed for each S^{2-} ion, making **Cu_2S** .

Binary Ionic Compounds – Worksheet #1

A. Write the formulas for the compounds formed from these elements. Remember, the cation is always written first.

- | | |
|---------------------------------|--------------------------------|
| 1. rubidium and iodine _____ | 5. sulfur and sodium _____ |
| 2. barium and chlorine _____ | 6. aluminum and oxygen _____ |
| 3. lithium and selenium _____ | 7. silver and phosphorus _____ |
| 4. nitrogen and magnesium _____ | 8. fluorine and zinc _____ |

B. Write the names for these binary ionic compounds.

- | | |
|-----------------------------------|-----------------------------------|
| 9. Cs_2S _____ | 16. FeF_3 _____ |
| 10. BaO _____ | 17. Mg_3N_2 _____ |
| 11. AlI_3 _____ | 18. Ni_3P_2 _____ |
| 12. MnO_2 _____ | 19. UO_2 _____ |
| 13. Tc_3P_4 _____ | 20. HF _____ |
| 14. CdBr_2 _____ | 21. CoN _____ |
| 15. NaCl _____ | 22. K_2S _____ |

C. Write the formulas for these binary ionic compounds.

- | | |
|---------------------------------|--------------------------------|
| 23. rubidium sulfide _____ | 30. europium(II) nitride _____ |
| 24. mercury(II) oxide _____ | 31. cesium phosphide _____ |
| 25. calcium nitride _____ | 32. lead(II) chloride _____ |
| 26. zinc bromide _____ | 33. cadmium oxide _____ |
| 27. uranium(VI) fluoride _____ | 34. tin(IV) fluoride _____ |
| 28. silver phosphide _____ | 35. iron(II) oxide _____ |
| 29. platinum(II) selenide _____ | 36. iron(III) oxide _____ |

Binary Ionic Compounds – Worksheet #2

If the name of the compound is given, write the formula. If the formula of the compound is given, write the name.

1. KBr _____
2. V_2O_5 _____
3. cobalt(III) oxide _____
4. barium phosphide _____
5. cadmium nitride _____
6. Cu_3P _____
7. Ag_2S _____
8. Sn_3N_4 _____
9. radium iodide _____
10. beryllium selenide _____
11. Fe_2S_3 _____
12. SrO _____
13. $CrCl_2$ _____
14. mercury(II) fluoride _____
15. lead(IV) bromide _____
16. CuSe _____
17. FeP _____
18. lithium oxide _____
19. cobalt(III) fluoride _____
20. CdI_2 _____

Ternary Ionic Compounds - Worksheet

If the name of the compound is given, write the formula. If the formula of the compound is given, write the name.

1. calcium nitrite _____
2. BaSO_4 _____
3. silver acetate _____
4. SrSO_3 _____
5. nickel(II) phosphate _____
6. Na_2CO_3 _____
7. LiHCO_3 _____
8. ammonium phosphate _____
9. $\text{Be}(\text{ClO})_2$ _____
10. aluminum oxalate _____
11. rubidium dichromate _____
12. KHSO_3 _____
13. calcium hydroxide _____
14. manganese(II) silicate _____
15. HCN _____
16. cesium hydrogen sulfate _____
17. $\text{Ti}(\text{OH})_4$ _____
18. ammonium chloride _____
19. $\text{Ca}(\text{ClO}_3)_2$ _____
20. rubidium cyanate _____
21. copper(II) sulfate _____
22. CuCl _____
23. iron(II) arsenate _____
24. NH_4OH _____

Latin Nomenclature

Some ions have Latin names for them that are commonly used. When there are two possible charges for similar ions, the lower of the two is the “-ous” ion and the higher of the two is the “-ic” ion.

For your information, the Latin roots for the name of the metals are: Cu “cupr”, Fe “ferr”, Cr “chrom”, Sn “stann”, Co “cobalt”, and Pb “plumb”.

Cu^+ - copper(I) ion – cuprous ion

Cu^{2+} - copper(II) ion – cupric ion

Fe^{2+} - iron(II) ion – ferrous ion

Fe^{3+} - iron(III) ion - ferric ion

Cr^{2+} - chromium(II) ion – chromous ion

Cr^{3+} - chromium(III) ion – chromic ion

Sn^{2+} - tin(II) ion – stannous ion

Sn^{4+} - tin(IV) ion – stannic ion

Pb^{2+} - lead(II) ion – plumbous ion

Pb^{4+} - lead(IV) ion – plumbic ion

You must learn all of these Latin names. Remember the lower of the two is “ous” and the higher of the two is “ic”.

Name the following compounds with both methods:

1. Fe_2O_3 iron(III) oxide ferric oxide

2. Cu_2O _____ _____

3. SnO_2 _____ _____

4. PbO _____ _____

Hydrate Nomenclature

hydrate – a compound that releases water when heated

Example: $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

Cu	SO_4	▪	5	H_2O
copper(II)	sulfate	part of the compound	penta	hydrate

Example: $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ magnesium sulfate heptahydrate

Example: aluminum chloride hexahydrate $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$

Name the following hydrates:

1. $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ _____

2. $\text{Cd}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$ _____

3. $\text{ZnCl}_2 \cdot 6\text{H}_2\text{O}$ _____

4. $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$ _____

5. $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$ _____

mono-	one
di-	two
tri-	three
tetra-	four
penta-	five
hexa-	six
hepta-	seven
octa-	eight
nona-	nine
deca-	ten

Write the formulas for the following hydrates:

6. barium hydroxide octahydrate _____

7. sodium sulfate decahydrate _____

8. lithium chloride tetrahydrate _____

9. cobalt(II) chloride hexahydrate _____

10. sodium carbonate decahydrate _____

All Ionic Compounds - Worksheet

If the name of the compound is given, write the formula. If the formula of the compound is given, write the name.

1. lead(II) nitrate _____
2. sodium carbonate _____
3. potassium iodide _____
4. AgNO_3 _____
5. barium nitrate _____
6. Na_2SO_3 _____
7. potassium carbonate _____
8. sodium nitrate _____
9. barium acetate _____
10. hydrogen peroxide _____
11. potassium biphosphate _____
12. $\text{Ba}(\text{OH})_2$ _____
13. FeCl_3 (use the Latin nomenclature) _____
14. $\text{Fe}_3(\text{PO}_4)_2$ _____
15. aluminum sulfate _____
16. calcium hydroxide _____
17. tin(II) oxide _____
18. aluminum hydrogen carbonate _____
19. sodium perchlorate _____
20. copper(I) dichromate _____
21. potassium selenide _____
22. ZnS _____
23. stannic dichromate _____
24. sodium hydrogen phosphate _____
25. $\text{Fe}(\text{ClO})_2$ _____
26. $\text{NH}_4\text{CH}_3\text{COO}$ _____
27. copper(II) nitrate _____
28. potassium hypochlorite _____
29. iron(III) chromate _____
30. Ag_2SO_4 _____

****THIS MAY BE GRADED FOR CORRECTNESS****

Molecular Compound Nomenclature

Molecular compounds are composed of individually covalently bonded atoms. The simplest unit of a molecular compound is called a “molecule”. These compounds are composed of all nonmetals. They are sometimes called *covalent* compounds.

One system for naming these compounds is based on the use of prefixes.

mono-	one
di-	two
tri-	three
tetra-	four
penta-	five
hexa-	six
hepta-	seven
octa-	eight
nona-	nine
deca-	ten

When naming the molecular compound, the prefix that indicates the number of each atom is placed before the atom in the name. All molecular compounds end in “-ide”.

Example: N_2O IS dinitrogen monoxide

NOT dinitride monoxide (the first element does not end in “-ide”)

NOT dinitrogen monoxygen (the second element should end in “-ide”)

NOT dinitrogen monogoxide (often the “o” is dropped before a vowel)

Also, mono is not used to indicate one atom if it applies to the first atom.

Example: CO IS carbon monoxide

NOT monocarbon monoxide (no mono necessary to indicate the first)

NOT carbon oxide (the mono is necessary to indicate the number of O)

NOT carbon monogoxide (often the “o” is dropped before a vowel)

Molecular Compounds - Worksheet

If the name of the compound is given, write the formula. If the formula is given, write the name.

1. CF_4 _____
2. N_2O_5 _____
3. CS_2 _____
4. SO_3 _____
5. P_4O_8 _____
6. iodine tribromide _____
7. chlorine dioxide _____
8. sulfur hexafluoride _____
9. difluorine octachloride _____
10. tribromine nonatelluride _____
11. H_2O _____
12. P_2S_4 _____
13. N_2O_4 _____
14. XeF_4 _____
15. SI_4 _____
16. carbon dioxide _____
17. trinitrogen hexabromide _____
18. diiodine heptaselenide _____
19. CO _____
20. dicarbon octafluoride _____
21. P_4O_{10} _____
22. Si_3N_4 _____
23. Cl_2S_7 _____
24. NBr_5 _____
25. phosphorus trichloride _____
26. PI_3 _____
27. disulfur trioxide _____
28. PCl_5 _____
29. diiodine dichloride _____
30. dinitrogen monoxide _____
31. I_4O_9 _____
32. dihydrogen monoxide _____

Acid Nomenclature

If a hydrogen atom, H, ($1p^+$, $1e^-$, $0n^0$) loses an electron to become a hydrogen ion, H^+ , what is left? Only a proton is left. So, sometimes, a proton is written as H^+ .

Often, an acid is referred to as a “proton donor”. If something is a proton donor, it has an H^+ in the formula to donate to another substance during a chemical reaction.

There are many definitions of acids. For naming purposes, we will only be naming acids that begin with “H” in the formula.

Acids are said to be “aqueous”. Aqueous means something is dissolved in water. It is written as (aq), although acids are not always written with the (aq).

binary acid – an acid that contains only two different elements: hydrogen and one of the more electronegative elements

Example: $HCl(aq)$ – hydrochloric acid

oxyacid – an acid that contains hydrogen, oxygen and a third element, usually a nonmetal.

Example: $HNO_3(aq)$ – nitric acid

NAMING RULES: Since all the acids we are naming begin with a hydrogen ion, they are differentiated by the anion in the formula. If the anion is identified, then the acid can be named.

-ide ions hydrostemic acid Ex: $HCl(aq)$ Cl^- chloride ion hydrochloric acid

-ite ions stemous acid Ex: $HClO_2(aq)$ ClO_2^- chlorite ion chlorous acid

-ate ions stemic acid Ex: $HClO_3(aq)$ ClO_3^- chlorate ion chloric acid

Sulfur and phosphorus are exceptions. For sulfur you place sulfur in the blank. For phosphorus, you place phosphor in the blank.

When taking the name and writing the formula, be sure to combine the appropriate amount of H^+ with the anion to make a neutral compound.

Example: phosphoric acid phosphor-ic acid is from the phosph-ate ion, PO_4^{3-}
Therefore the formula is $H_3PO_4(aq)$.

So, if the citrate ion is $C_6H_5O_7^{3-}$, what is the formula for citric acid?

More Acids...

monoprotic acid -

diprotic acid -

triprotic acid -

polyprotic acid -

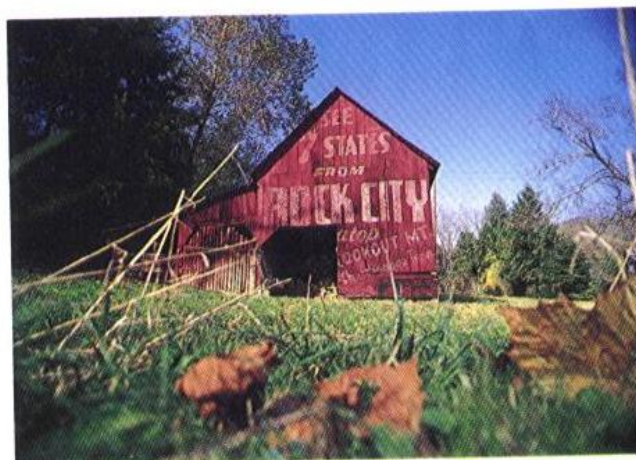
APPLICATIONS OF CHEMISTRY

Chemical Nomenclature in Daily Life

You may have encountered many simple, yet interesting chemical compounds for years without being aware of them. For example, did you know that barns are red today because historically they were once preserved with rust (Fe_2O_3)? (See photo on right.) In the mid-nineteenth century, American farmers made an inexpensive, yet long-lasting barn preservative by mixing rust-iron(III) oxide, skim milk, linseed oil, and lime (CaO , calcium oxide). By the late nineteenth century, red had become the traditional color for barns. Barns are no longer preserved with rust and the red color is the result of red paint pigments. But historically, they are red because of rust.

Another use of chemicals in everyday life can be found in the supermarket. What keeps meat at the supermarket looking fresh? The color, flavor, and texture of meats depend on additives such as sodium nitrite (NaNO_2) and sodium nitrate (NaNO_3). Sodium nitrate also acts as a preservative, and its use has virtually wiped out botulism food poisoning. Despite this, the Food and Drug Administration has been trying to limit the concentration of these two chemicals because excessive amounts may be harmful. Food processors add iron(II) sulfate (FeSO_4) and iron(III) phosphate (FePO_4) to many breads and cereals as a dietary source of iron.

You probably think that the only chemical in table salt is sodium chloride. But there can be any of 20 other chemicals in table salt. Sodium chloride is the primary ingredient, but others include potassium iodide (KI), which is a nutrient. Sodium hydrogen carbonate (NaHCO_3), sodium carbonate (Na_2CO_3), calcium hydroxide [$\text{Ca}(\text{OH})_2$], and sodium monohydrogen phosphate (Na_2HPO_4) are also present to help stabilize the potassium



iodide. Magnesium carbonate (MgCO_3), calcium hydrogen phosphate (CaHPO_4), calcium phosphate [$\text{Ca}_3(\text{PO}_4)_2$], and calcium carbonate (CaCO_3) are present as desiccants (drying agents) to help keep the salt crystals from fusing.

Some other chemicals that you may find around your own home include SiO_2 (silicon dioxide, sand), NaOH (sodium hydroxide, lye, used in drain cleaners), and H_2SO_4 (sulfuric acid, battery acid). Also common are CuSO_4 [copper(II) sulfate], an algicide, HCl (hydrochloric acid, called muriatic acid, used to clean masonry and bricks), and NaClO (sodium hypochlorite, used in liquid laundry bleach).

Acids - Worksheet

If the name of the acid is given, write the formula. If the formula of the acid is given, write the name.

1. hydrocyanic acid _____
2. dichromic acid _____
3. hydrobromic acid _____
4. nitrous acid _____
5. sulfuric acid _____
6. $\text{H}_2\text{SiO}_3(\text{aq})$ _____
7. $\text{HF}(\text{aq})$ _____
8. $\text{H}_3\text{PO}_3(\text{aq})$ _____
9. $\text{H}_2\text{CO}_3(\text{aq})$ _____
10. $\text{H}_2\text{S}(\text{aq})$ _____
11. acetic acid _____
12. sulfurous acid _____
13. perchloric acid _____
14. hydroselenic acid _____
15. carbonic acid _____
16. $\text{HClO}(\text{aq})$ _____
17. $\text{HClO}_2(\text{aq})$ _____
18. $\text{H}_2\text{C}_2\text{O}_4(\text{aq})$ _____
19. $\text{H}_3\text{P}(\text{aq})$ _____
20. $\text{HMnO}_4(\text{aq})$ _____
21. hydrochloric acid _____
22. chromic acid _____
23. phosphoric acid _____
24. $\text{HCl}(\text{g})$ _____

Chapter Review of All Naming

If the name of the substance is given, write the formula. If the formula is given, write the name.

1. cadmium nitrate _____
2. chromate ion _____
3. dinitrogen monoxide _____
4. potassium bromide _____
5. nitrous acid _____
6. HPO_4^{2-} _____
7. PCl_3 _____
8. V_2O_5 _____
9. HClO_4 _____
10. BaSO_4 _____
11. sulfur hexafluoride _____
12. cuprous oxide _____
13. sulfuric acid _____
14. hydrogen peroxide _____
15. carbonate ion _____
16. BaO _____
17. HClO_2 _____
18. SrSO_3 _____
19. Fe _____
20. acetic acid _____
21. CBr_4 _____
22. hypochlorite ion _____
23. $(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ _____
24. calcium hydroxide _____
25. Na_2CO_3 _____
26. Cu_3P _____
27. HI _____
28. lead(II) acetate _____

Polyatomic Ion Practice

(Some of these, you did not have to memorize.)

NO_2^-	_____	hydroxide ion	_____
CO_3^{2-}	_____	hydrogen sulfate ion	_____
$\text{B}_4\text{O}_7^{2-}$	_____	hydrogen phosphate ion	_____
HSO_3^-	_____	hydrogen carbonate ion	_____
SCN^-	_____	chlorate ion	_____
H_2PO_4^-	_____	ammonium ion	_____
$\text{C}_2\text{O}_4^{2-}$	_____	perchlorate ion	_____
PO_3^{3-}	_____	hypochlorite ion	_____
H_3O^+	_____	nitrate ion	_____
AsO_4^{3-}	_____	cyanide ion	_____
NH_4^+	_____	sulfate ion	_____
MnO_4^-	_____	sulfite ion	_____
BO_3^{3-}	_____	chromate ion	_____
PO_4^{3-}	_____	chlorite ion	_____
$\text{Cr}_2\text{O}_7^{2-}$	_____	acetate ion	_____
O_2^{2-}	_____	thiosulfate ion	_____
I_3^-	_____	cyanate ion	_____